Slip 09

Q1) A Explain the key generation, encryption, and decryption steps of the RSA algorithm with mathematical formulas

1. Introduction

RSA (Rivest–Shamir–Adleman) is an asymmetric encryption algorithm that uses two keys: a public key for encryption and a private key for decryption. It is widely used to protect information in digital communication.

2. Key Generation

Key generation involves the following steps:

Step 1: Select two prime numbers

Choose two prime numbers p and q.

Example:

$$p = 3, q = 11$$

Step 2: Calculate n

Multiply the two prime numbers.

$$n = p \times q$$

$$n = 3 \times 11 = 33$$

Step 3: Calculate Euler's totient function

$$\phi(n) = (p - 1)(q - 1)$$

$$\phi(n) = (3 - 1)(11 - 1) = 20$$

Step 4: Select the public exponent e

Choose e such that:

$$1 < e < \phi(n)$$

and

$$\gcd(e, \phi(n)) = 1$$

Choose $e = 3$, because $\gcd(3, 20) = 1$.

Step 5: Calculate the private exponent d

Find d such that:

\[(d\times e)\bmod\phi(n)=1 \]

Here, $d = 7$, because:

\[(7\times3)\bmod20=1 \]

Therefore:

- Public key: $(e, n) = (3, 33)$
- Private key: $(d, n) = (7, 33)$

3. Encryption

The sender uses the receiver's public key to encrypt the message.

Formula:

\[C=M^e\bmod n \]

Where:

- M = numerical representation of plaintext
- e = public exponent
- n = modulus
- C = ciphertext

Suppose $M = 4$.

\[C=4^3\bmod33 \]
\[C=64\bmod33=\boxed{31} \]

Therefore, the encrypted message is 31.

4. Decryption

The receiver uses the private key to recover the original message.

Formula:

\[M=C^d\bmod n \]

Using $C = 31$, $d = 7$, and $n = 33$:

\[M=31^7\bmod33 \]

Since \(31^2\bmod33=4\), we get:

\[31^7\bmod33=\boxed{4} \]

Therefore, the decrypted message is 4, which matches the original message.

5. RSA Process Diagram

Key Generation

Public key: (3, 33)

Private key: (7, 33)

Plaintext: 4

Encryption using public key

\(C=4^3\bmod33=31\)

Ciphertext: 31

Decryption using private key

\(M=31^7\bmod33=4\)

Original plaintext: 4

6. Conclusion

RSA provides secure communication by using a public key for encryption and a private key for decryption. Its security depends on the difficulty of factoring large numbers. The small numbers used in this example are only for educational purposes; real RSA requires much larger keys and secure padding schemes.

Q2) B Write a Python program using PyCryptodome to encrypt and decrypt the message "HELLO WORLD" using AES with a predefined 16-byte key

```
from Crypto.Cipher import AES
from Crypto.Util.Padding import pad

# Message
message = "HELLO WORLD"

# AES key (16 bytes = 128-bit key)
key = b"1234567890123456"

# Create AES cipher object
cipher = AES.new(key, AES.MODE_ECB)

# Convert message into bytes and add padding
data = pad(message.encode(), AES.block_size)

# Encrypt the message
encrypted = cipher.encrypt(data)

print("Original Message :", message)
print("Encrypted Message:", encrypted.hex())
```

OR

Q2) B Write a Python program to encrypt user input text using AES encryption

```
from Crypto.Cipher import AES
from Crypto.Random import get_random_bytes
from Crypto.Util.Padding import pad

key = get_random_bytes(16)
```

```
cipher = AES.new(key, AES.MODE_ECB)

text = input("Enter text: ")
data = pad(text.encode(), 16)

encrypted = cipher.encrypt(data)

print("Encrypted text:", encrypted.hex())
```